Automated Detection and Classification of Solar Radio Bursts

Heliophysics: Radio Instrumentation Research Team - Software

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UNIVERSITY OF MICHIGAN

SunRISE Sun Radio Interferometer Space Experiment

CLASP CLIMATE & SPACE Sciences & Engineering

Introduction

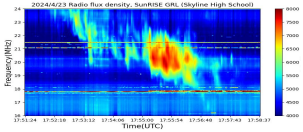
As part of the MDP Heliophysics research project, the software subteam is responsible for improving the quality and accessibility of the spectrograms output by our Ground Radio Lab (GRL) and Long Wavelength Array (LWA) antennas, as well as creating a system to automatically detect and classify Solar Radio Bursts (SRBs). The long term goals of this team are:

- Accurately detect and classify solar radio bursts of all types
- Improve public access to spectrogram visualization and data
- Maintain data uploading system at Marquette High School

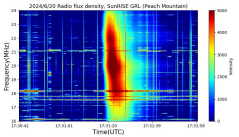
Background

Solar Radio Bursts are high-intensity emissions of electromagnetic waves by the Sun, which can be caused by a variety of solar events.

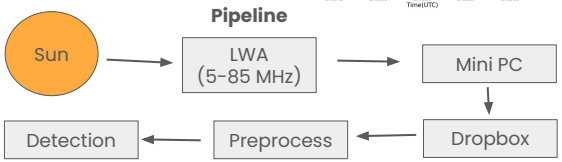
Type II



Type III



Pipeline



Data Collection & Generation

Our data comes from publically available datasets as well as our own data collection efforts in Marquette and on Peach Mountain. Below is the size of our current dataset used for ML training, and the flow of data from the LWA antenna to our Dropbox.

Class	Boxes	Windows containing
Type II	221	214
Type III	3,947	3,819
Type V	86	86
Any burst	4,254	4,047

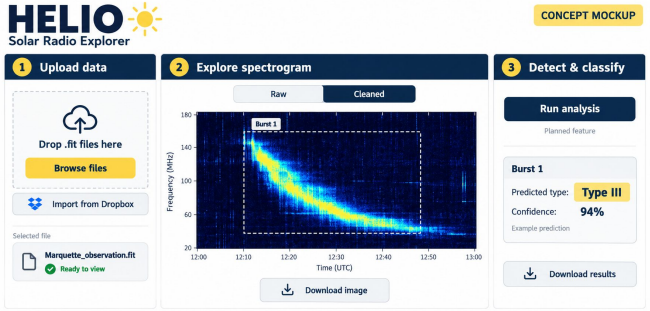
current labeled dataset size

Marquette HS Data Uploading Pipeline

1. LWA Antenna
2. e-CALLISTO Receiver
3. CALLISTO Software
4. .fit File creation
5. Schedules Daily Upload
6. Dropbox

Preprocessing

The preprocessing task group is developing a public platform to host our preprocessing and detection models to allow Marquette HS students to see their data visualizations.

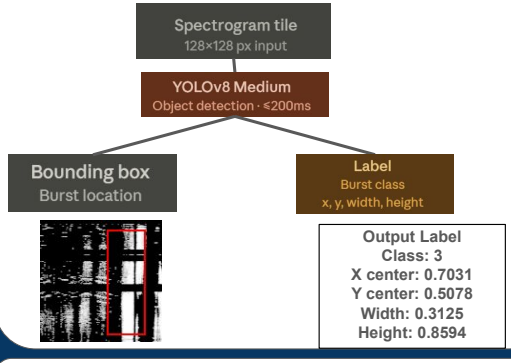


Initial website mockup for Marquette students to view spectrograms

Machine Learning

Machine Learning is employed in the *Detection and Classification* of solar bursts of different types (I-VII) from spectrograms. We use a **YOLO (You Only Look Once)** model for object detection.

- Real time solar burst detection
- Good spatial representation learning
- Bounding box localization of burst events
- Detection of multiple bursts in the same spectrogram



Future Steps

1. Create working website based on mock-up
2. Improve accuracy of detection model (>82%)
3. Implement classification model
4. Increase size of SRB dataset for training using Marquette and Peach Mountain Antenna Data